The Questions

1.

a)

i) Explain the difference between lightning and switching surges in power systems.

[4]

The students are expected to relate lightning is an external event to the system, it is high current short duration, need to characterise with large current and short duration, for switching it is an internal event mainly circuit opening and close following a fault – it is associated with large over voltage like in hundreds of kVs and hundreds of milliseconds. while current in lighting protection design for overhead conductor and power apparatus like transformer, switching surge dictates insulation design consideration of breaker and apparatus.

ii) How do these phenomena impact the operational stability of a power system ?

[4]

Lightning can give rise to temporary faults, this is a transient stability issues, rotor angle stability. switching also impacts transient stability it also has impact on voltage stability. Over voltage may lead to transformer saturation etc.

iii) What are stator transients in the context of power system stability ?

[2]

Stator transients are 50 Hz power frequency component in stator during disturbance. it comes from transformer EMF – they are purely electromagnetic in nature.

iv) How are these stator transients modelled for electromechanical stability studies involving synchronous machines ?

[2]

In transient stability or electromechanical stability study, these transients are ignored – however for torsional oscillations study they are included.

v) How are these stator transients modelled for stability studies involving asynchronous machines used for wind power applications ?

[2]

In asynchronous machines used for wind power applications with power converter, they are modelled as differential equations.

b)

i) What is electrical resonance in the context of power system operation ?

[4]

The student is expected to answer like this: electrical resonance is a phenomenon that involves interaction between energy storing elements like inductance and capacitances, usually when a sinusoidal voltage/current signal with a frequency matching the natural frequency involving inductance capacitance is present, the oscillating current and voltages are manifested in the path connecting these elements. The resistance is usually small, so the current is high. Usually the source of these signals are high power electronics switching, not filtered out completely from flowing into the networks where they meet inductance and capacitance. This has been a new trend in converter interfaced generation.

ii) Where are such resonance phenomena observed ?

[2]

They are observed in networks, in real and reactive power flow, node voltages etc.

ii) How sub-synchronous torsional interaction (SSTI) is different from electrical resonance ?

[3]

Students are expected to write the origin and manifestation of SSTI – involving network and turbine masses , for electrical resonance – they need to relate where it comes from – like power electronics converter control and its spread , like in the network – purely electrical is the point they need to know and state.

iii) How has more converter interfaced generation (CIG) in electric power systems influenced the degree of presence and impacts of both the above phenomena ?

[2]

The nature has evolved from electro mechanical to electrical where control is more taking central part in the problem . In all converter interfaced generation, voltage and power oscillations in the network is more dominated by converter , network interactions . SSTI is absent in Type 4 wind turbine , there is very little in type 3 wind turbine. In solar PV farm , there is no rotating element involved , none of it will be SSTI , if it is present , it will be purely electrical in nature.

[25 marks]

2.

a)

i) Some synchronous generators are of 2-pole while some are 4, 6 , 8 or even higher poles – Explain the reasons with reference to suitable applications.

[3]

The students need to state their understanding that steam turbine is very efficient at high speed – so thermal and nuclear based generator will have 2 poles. In hydro generation the speed depends on the water head and pressure, as such efficiency is not an issue , so large poles are required to meet the network frequency.

ii) Usually the stator of a synchronous generator is laminated whereas the rotor is solid. However, in many cases additional windings are housed in the rotors. What are the purposes of these windings and how do they impact the stability performance of the generator as well as that of the power system?

[2+2]

Stator has power frequency, so hysteresis and eddy losses are an issue in the stator. Lamination is must .

[2]

However , the rotor is DC excited, no speed or transformer voltage is induced at steady state – during transient voltage is induced in rotor body -which drives current to produce torque. Such torque is useful to dampen the over speed or oscillation – sometimes this is not enough, so it is necessary to have physical closed winding to produce an extra torque – that justifies having additional windings.

[2]

iii) Synchronous machines with larger air gap have better synchronisation capability. But in practice, a very small or minimum air-gap construction is adopted. Explain the reason behind this, and also narrate how synchronisation capability is ascertained in low air-gap machines.

[2+2]

Larger airgap has more space to store energy in its electromagnetic field, which is instrumental in synchronisation – this also gives rise to larger synchronising power co-efficient.
Lower gap machines are cheaper to make as overall volume is small, materials requirement is lesser. Usually, low synchronisation torque or capability is addressed through high gain fast acting excitation system control

b) A round rotor star connected synchronous generator is connected to the grid at 20 kV (line to line) voltage. The rated power of 500 MW, power factor is 0.85 (lagging). The synchronous reactance is 1.50 p.u. on its rated base of 588 MVA. The resistance is negligible. Find out:

i) the rated line current in kA

[2]

ii) the excitation (phase) voltage in kV

[3]

iii) load angle (δ) in degree

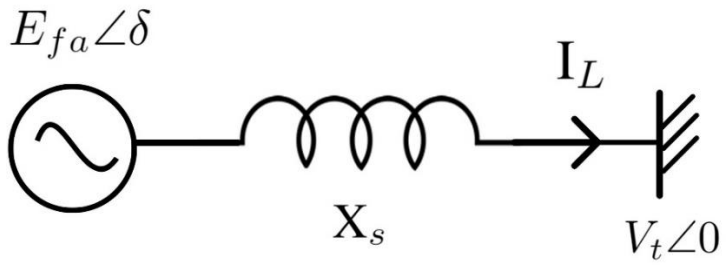
[2]

iv) steady state stability margin in degree

[1]

iv) total reactive power in MVAR

[3]



$V_t = 20\ kV, \quad P_{rated} = 500\ MW, \ \ Rated\ MVA = 588$
$Power\ factor = 0.85\ lag$
$I_L = \dfrac{MVA}{\sqrt{3}V_t} = 16974\ A$

$E_{fa} = V_t + j\sqrt{3}I_L X_s$
$Base\ Impedance = \dfrac{kV^2}{MVA}, \quad Power\ factor\ angle = -cos^{-1}0.85 = -31.78°$
$X_s\ in\ \Omega = X_s\ (p.u.)\dfrac{kV^2}{MVA} = 1.5 \times 0.6803 = 1.02045\ \Omega$
$E_{fa} = 43.96\ kV\ \angle 35.46°$
$E_{fa}/phase = 25.38\ kV$
$load\ angle\ delta = 35.46°$
$Stability\ Margin = 90° - 35.46° = 54.54°$
$Reactive\ Power = \dfrac{E_{fa}}{X_s}V_t\ cos\ \delta - \dfrac{{V_t}^2}{X_s} = 310 MVAr$

c) The excitation voltage is decreased by 20%

i) Find out the change in stability margin in degree with respect to the rated excitation voltage.

[3]

Real power does not change. So:

$$P = \frac{0.8E_{fa}V_t}{X_s} sin\,\delta'$$

$$\delta' = 46.5°$$

$$Stability\ Margin\ =\ 43.5°$$

[25 marks]

3.

a)

i) Why does one need to move to a rotational reference frame for synchronous machine dynamic modelling ?

[2]

The stationary reference frame in 3 phase result in time variant inductance between stator and rotor . In rotational reference frame that time variance is removed.

ii) How does a rotational reference frame help in voltage, current and power calculation in synchronous machine dynamic analysis ?

[3]

The students are expected to demonstrate their understandings that stator self and mutual inductance is time varying because of the rotation of the rotor. This makes the voltage and current relationship very complicated to analyse. In rotational reference frame everything is DC at steady state and at synchronous speed which is easy to analyse.

iii) Why, in a multi-machine power system context, the variables in rotational reference frame need further transformation ?

[3]

Usually, terminal voltage of a generator is taken as reference in Park or other rotational transformation – for single machine infinite bus – this is fine. In reality, a generator is part of the entire power system where many more generators are connected. This requires all of the individual terminal reference voltages to be expressed with respect to one common reference voltage. That reference usually comes from power flow solution. This requires reference adjustment through another transformation called rotational to Network transformation.

iv) In lagging power factor operation, the transformed current and voltage in d-axis oppose the field current and exciation voltage. Explain the reasons.

[4]

In lagging power factor operation, the stator magnetic field is demagnetising in nature. This is reflected in the sign of voltage and current of d-axis when field current and field voltage are taken as positive. That is also the reason why the generator is over excited.

v) What impact will such a lagging power factor operation have on the generator small signal as well as large signal (transient) stability ?

[2+2]

Lagging power factor has negative impact on small signal stability as Excitation system with fast AVR may introduce negative damping at low frequency range. [2]

In transient stability it is beneficial as it will have lower load angle for a given MW output compared to that with leading power factor operation. It is the high gain and fast acting AVR action that provides improvided transient stability

[2]

b)

i) What are the basic functional and constructional differences between the Type 3 (DFIG) and Type 4 (full converter) wind turbine generators (WTGs) ?

[4]

Type 3 will have rotor power recovery scheme; it will have power generation capacity in wider wind velocity range both sub and super synchronous range – power flows to the network through both stator and rotor. It also requires gear. It will have power converters on the rotor which is lesser in capacity compared to that of Type 4 for the same rating.
Type 4 is single power path through stator. It can operate at much lower generator rotor speed. The converter must be of total rated capacity unlike Type3 which is about 30% of the total power capacity. It has large number of poles , so low speed and larger diameter is permitted , it will have bigger nacelle though.

ii) Which of the active stall and active pitch control in modern WTGs is the preferred choice and why ?

[1+2]

Active pitch control is preferred as it lets the wind through by adjusting angle of attack . The active stall adjust the angle of attack in a way it counter attack the wind – this stress and shear on the blades are more impacting the operating life

iii) What is the importance of optimal tip speed ratio (TSR) in the context of WTG operation ?

[2]

Below rated wind speed, the power conversion efficiency (Cp) from wind to mechanical form is dependent on the turbine speed to wind velocity ratio in a very complex way – it is found that as that ratio changes the Cp increasing nonlinearly to a maximum and then again starts falling. It is the tip speed ratio at which the power conversion is highest. It is important to have that optimal value for optimal power captured at subrated wind speed.

[25 marks]

4.

a)

i) Define co-efficient Cp in wind power conversion.

[2]

Cp, is the 'coefficient of performance', or the percent of maximum possible energy that can be harvested from the air hitting the wind turbine. The maximum value is 16/27 . This happens when the wind slows down by precisely two-thirds as it passes the turbine. At maximum efficiency, the column of air fans out over a 3x larger area.

ii) What design and operating parameters influence Cp ?

[2]

Angle of attack, wind velocity , turbine speed and pitch angle influences Cp.

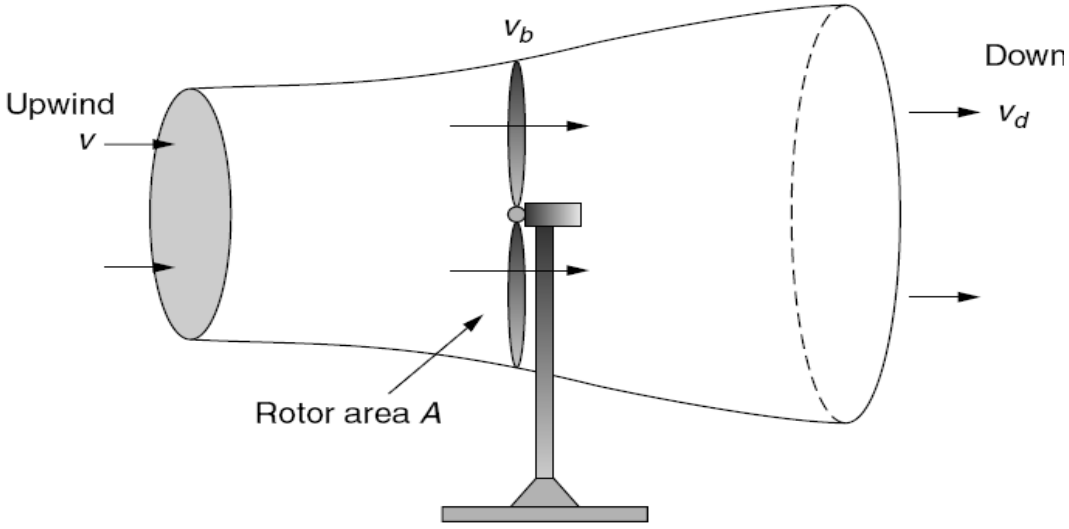
iii) Starting from the first principle of aerodynamics, and including the assumptions, establish that the Betz limit is 16/27.

[7]

This one has been taught in the class – the student can derived from the first principle and is available in their lecture notes:

1. Betz limit derivation

Look at the figure below



Suppose

the rotor swept area is A, the upwind speed is v, the downwind speed is v_d, the average wind speed at blades is $v_b = \frac{v+v_d}{2}$, and the mass flow of air is $\dot{m}$

[1 mark]

The power extracted by the blade is

$P_b = \frac{1}{2}\dot{m}(v^2 - v_d^2)$ [1]

the mass flow of air can be expressed as

$$\dot{m} = \rho A v_b$$

Substituting $\dot{m}$ into P_b yields

$$P_b = \frac{1}{2}\rho A\left(\frac{v + v_d}{2}\right)(v^2 - v_d^2)$$

Assume $\lambda = \frac{v_d}{v}$, the expression of P_b becomes

$$P_b = \frac{1}{2}\rho A\left(\frac{v + \lambda v}{2}\right)(v^2 - \lambda^2 v^2)$$

$$P_b = \frac{1}{2}\rho A v^3\left[\frac{1}{2}(1+\lambda)(1-\lambda^2)\right]$$

$$P_b = \frac{1}{2}\rho A v^3 C_p$$

with $C_p$is equal to turbine efficiency.[2]

In order to find the maximum value, take the derivative of C_p with respect to λ and set this to zero

$$\frac{dC_p}{d\lambda} = \frac{1}{2}[(1+\lambda)(-2\lambda) + (1-\lambda^2)] = 0$$

After simple calculation, the solution is $\lambda = \frac{1}{3}$. Substituting back the value of λ into C_p results

$$C_p = \frac{1}{2}\left(1 + \frac{1}{3}\right)\left(1 - \frac{1}{3^2}\right) = \frac{16}{27}$$

[3]

b)

i) What is the fundamental difference between pitch and stall control ?

[2]

Active stall and pitch control both require the mechanical manoeuvring of the blade angle. In one cases it faces the wind more which is stalled control and in pitch control it changes the angle in a way , it faces less wind

iv) How can WTG offer frequency response service at subrated wind speed?

[3]

In the sub rated wind speed there will be a maximum power point after adjusting to optimal tip speed ratio. In order to have frequency response service from WTG at that speed range , there has to be reserve : ie power has to be set at 80% of MPPT keeping 20% power available to tap when needed . In reality this is not spinning reserve as wind energy not captured is wind energy lost – but that is fine as there is no price for wind and the WTG owner will be paid for this capability ,so there is no losses of revenue

iv) How is it possible to have net power production from Type 3 WTG below the synchronous speed ?

the control in the rotor and stator side conveter in DFIG can produce bi directional current . So at below synchronous speed current can enter the rotor , this is slip frequency current produce rotating MMF along the direction of rotor

speed – thus resultant speed is synchronous , the stator will have power frequency voltage at a wind speed below the wind speed .

[3]

v) For Type 4 WTG, why is a gearbox not mandatory ?

[2]

Type 4 uses a full conveter , whatever inputs speed the rotor roates with the help of large number of poles a descent AC voltage with 20-40 Hz is generated , which is then rectified to DC and invetered at network frequency. So gear box is not required as rotor speed regulation is not needed unlike Type 1, 2 and 3.

vi) Why are turbine blades not controlled to track fast and random wind speed variation ?

[2]

The blades are very heavy and it needs huge power and effort to move, it also moves slowly . The bearing system is subjected to force. For randon fluctuation and temporary changes , it is better to keep the blade unpitched to keep it operating life unaccelerated.

vii) Why in a wind farm, cables are never loaded beyond less than half of their rated capacity even at the rated wind speed ?

[2]

Cables are generally 3-4 times more expensive than electrical overhead conductors. Underground cables used in wind farm application is of either copper or aluminium. They have an operating life 70 years compared 25 years from windfarm. So, one needs three life upgradations of WF over the entire life of the cables. As wind farm renewal takes place, it will host more higher capacity efficient wind turbine, thus at the next round of wind farm renewal, the cable does not need to be upgraded as it has 70 years life and can easily carry increased power. Replacement of UG cables is very expensive and environment unfriendly option.

[25 marks]